

ADITYA KARALE

Senior Platform Engineer | SRE | DevOps Engineer | Observability Specialist

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PROFESSIONAL SUMMARY

Platform Engineer and SRE with 4+ years at Jio Platforms Limited, specialising in production Kubernetes lifecycle management (v1.18 to v1.34, 16+ upgrade cycles, zero critical downtime), enterprise-scale ELK observability (42-node cluster, 100+ TB, ~300K log events/minute, 3,500+ APM services at 99.9%+ availability), and Azure cloud infrastructure (AZ-104 and AZ-400 certified). Designed and built OCCNXT, a production Operation Command Center adopted by 14+ enterprise applications with real-time health monitoring across Kubernetes, Kafka, databases, SAP, and external services for multiple users. Pioneered AI-powered SRE tooling using private LLMs, RAG, and MCP enabling on-call engineers to query live platform telemetry via natural language. Targeting a Senior Platform Engineer, SRE, or Observability Specialist role to drive large-scale platform reliability and engineering excellence.

TECHNICAL SKILLS

Cloud & Azure: Microsoft Azure, AKS, ACR, Azure VNet, Subnets, Private DNS, NAT Gateway, Azure Bastion, Azure DevOps, ARM Templates, GCP

Containers & Orchestration: Kubernetes (v1.18–v1.34), AKS, Docker, Helm, KEDA, Descheduler, HPA, Weave, Cilium, eBPF, RBAC, OPA/Gatekeeper, CRI-O

Observability & Monitoring: Elasticsearch, Logstash, Kibana, Elastic APM, ELK Stack (v7.4–v9.2), Metricbeat, Filebeat, Prometheus, Grafana, Custom Logstash Pipelines

CI/CD & DevOps: Azure DevOps Pipelines, GitLab CI/CD, GitHub, Fortify SCA, Istio, CRI-O

Infrastructure & IaC: Terraform, Ansible, ARM Templates, Bash, Shell Scripting, Python

Backup & Disaster Recovery: Velero, Kubernetes Backup & Restore, Filesystem Replication

Databases & Messaging: MySQL, MongoDB, Oracle, Redis, Kafka, Milvus

AI & Platform Tooling: Ollama, RAG Architecture, MCP (Model Context Protocol), Vector Databases, AI SRE Chatbot, LLM Integration

OS & Systems: Linux, RHEL, Ubuntu, CentOS, Windows Server

PROFESSIONAL EXPERIENCE

Assistant Manager – Platform & DevOps Engineering | Jio Platforms Limited

Navi Mumbai, Maharashtra | November 2021 – Present

Kubernetes Platform Engineering & Lifecycle Management

- **Owned end-to-end Kubernetes lifecycle** across multi-environment production and non-production clusters, sustaining 99.9%+ uptime and workload stability throughout 4+ years of continuous operation.
- **Executed 16+ structured upgrade cycles** from v1.18 to v1.34, maintaining full API compatibility and zero critical downtime; coordinated parallel version tracks across DEV, QA, PPD, and PROD environments.
- **Led CNI migration** from Weave to Cilium (eBPF-based networking) across DEV, QA, and PPD, improving network policy enforcement, traffic routing, and kernel-level observability.
- **Deployed Kubernetes Descheduler** with taint- and toleration-aware eviction policies across multiple environments, reducing cluster resource waste by ~20% through automated workload rebalancing.
- **Implemented Velero-based backup and disaster recovery**, enabling point-in-time cluster restores with RTO under 30 minutes for critical namespaces.
- **Directed CentOS-to-Ubuntu node migration** across DEV, QA, PPD, and PROD using Velero strategy for lower environments and filesystem replication for PROD, achieving zero workload disruption.
- **Migrated container runtimes** from Docker to containerd, then to CRI-O; installed Kubernetes v1.27.1 with CRI-O on GCP servers as part of runtime modernisation.
- **Upgraded Istio service mesh** from v1.13.2 to v1.17.2 across all environments, improving traffic management, observability, and mTLS enforcement.
- **Automated kubeconfig admin certificate rotation** via pipeline-based generation and replacement, eliminating manual certificate expiry incidents across clusters.

OCCNXT — Operation Command Center (Flagship Project, Dec 2025 – Mar 2026)

- **Architected and delivered OCCNXT** (occnxt.ril.com), a production SRE command center providing unified real-time visibility across Core Technical Services, Application Services, User Journey Services, External Services, Vendor Applications, and Infra Usage.
- **Onboarded 14+ enterprise applications** including ProcureNXT, CustomerFirst, FR8First, TrackFirst, PeopleFirst, JioFace, IRCMS, and Finance Services, with per-application configuration, approval workflows, and tenant-level RBAC for multi users.
- **Built embedded SRE Check engine** per technology (Kafka, Kubernetes, Redis, MongoDB, MySQL, Oracle, Elasticsearch), enabling one-click automated health checks with structured pass/fail diagnostics — e.g., Kafka consumer lag detection across 43 consumer groups and 51 topics.
- **Implemented node-level Infra Usage view** delivering CPU, memory, load average, and disk metrics per technology cluster with configurable warning (70%) and unhealthy (85%) thresholds.

- **Engineered AI SRE chatbot** using Ollama (private LLM) with MCP calls, enabling engineers to query live platform telemetry in natural language and receive LLM-generated health summaries across the full application stack.
- **Integrated RAG over SRE runbooks** for Kafka, Kubernetes, MongoDB, Redis, and more enabling on-call engineers to retrieve contextual troubleshooting guidance directly within the OCCNXT interface.

Large-Scale ELK Observability Platform

- **Operated 42-node distributed ELK cluster** with Hot-Warm tiered architecture across 100+ TB, processing ~300K log events/minute, supporting 3,500+ APM services and 1,300+ Metricbeat/Filebeat integrations at 99.9%+ availability.
- **Led ELK upgrade** from v7.4 to v9.2 across production via rolling upgrades with zero downtime; migrated underlying infrastructure from CentOS to RHEL for enterprise stability and vendor compliance.
- **Engineered custom Logstash pipelines** including dedicated pipelines for WSO2 API Gateway log analysis across all environments, reducing parse latency and improving log fidelity.
- **Built Kibana dashboards for real-time SLA monitoring**; integrated Elastic APM for distributed latency tracing, reducing P1 MTTR by ~35%.

Azure Cloud & AKS Platform Engineering

- **Supported enterprise AKS workloads**, configuring ACR, VNet segmentation, subnets, Private DNS, NAT Gateway, and Azure Bastion for secure private cluster and container registry connectivity.
- **Automated Azure infrastructure provisioning** using Terraform and ARM Templates; built Azure DevOps pipelines integrating AKS and ACR into deployment workflows.
- **Diagnosed and resolved DNS resolution and container registry connectivity issues** across AKS environments, improving platform reliability for dependent application teams.
- **Upgraded Docker** from v18.09.0 to v23.0.06 on Azure Build Servers, improving container build security and runtime performance.

KEDA, CI/CD, Automation & Monitoring

- **Designed and rolled out KEDA** for event-driven autoscaling, integrating Prometheus metrics with ScaledObjects, HPA behaviour policies, polling intervals, and cooldown periods; validated via Grafana and Helm-deployed ScaledObjects.
- **Built and maintained CI/CD pipelines** using Azure DevOps and GitLab CI/CD for containerised microservices; integrated Fortify SCA static code analysis for automated security scanning ahead of production promotion.
- **Developed automated email reporting** covering component status across the platform, replacing manual stakeholder status checks with scheduled structured reports.
- **Maintained Prometheus metrics pipelines and Grafana dashboards** for Kubernetes resource utilisation, application health, and autoscaling monitoring; established alerting for node readiness and critical service downtime.

EDUCATION

Master of Technology (MTech) – Software Systems | Birla Institute of Technology and Science (BITS), Pilani | 2022 – 2026

Bachelor of Science (B.Sc.) – Computer Science | Fergusson College, Pune | 2018 – 2021

KEY ACHIEVEMENTS

- **Star Performer Award** – recognised by senior leadership at Jio Platforms Limited for stellar performance and valuable contributions.
- **Kubernetes Reliability**: drove 16+ upgrade cycles (v1.18 to v1.34) over 4 years with zero critical downtime across all environments.
- **ELK Observability Scale**: operated a 42-node cluster spanning 100+ TB, ~300K log events/minute, 3,500+ APM services at 99.9%+ availability.
- **Incident Response**: reduced P1 MTTR by ~35% through Elastic APM integration and AI SRE chatbot tooling within OCCNXT.
- **Platform Engineering**: built and launched OCCNXT — onboarding 14+ enterprise applications, real-time SRE visibility, AI-powered diagnostics, and RBAC-governed multi-tenant access for multiple users.
- **Resource Optimisation**: reduced cluster resource waste by ~20% through Descheduler implementation and automated workload rebalancing.

CERTIFICATIONS, RECOGNITION & DEVELOPMENT

- **Microsoft Certified: Azure Administrator Associate (AZ-104)**
- **Microsoft Certified: DevOps Engineer Expert (AZ-400)**
- **Jio Certified Google Cloud Platform Professional**
- **Jio Certified AI & ML Foundation**
- **Kubernetes Specializations – LinkedIn Learning**
- **Elastic Stack – LinkedIn Learning**

Core Competencies: Stakeholder Communication • Agile / Sprint Collaboration • Problem-Solving • Cross-Functional Team Support • End-to-End Ownership • Knowledge Sharing & Mentorship

Languages: English (Professional) • Marathi (Native) • Hindi (Fluent)